CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: Assessing the Performance Gain of Stacked Ensemble Models Over Traditional Machine Learning Methods for Emergency Department Triage in a Caribbean Emergency Departments

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Statement of the Problem:

At Mercer ED, limited resuscitation bed capacity places pressure on timely and accurate patient triage. The department operates in a Caribbean hospital, sharing laboratory, radiology and pharmacy services with other departments. Previous studies demonstrate that AI-integrated triage systems can improve triage performance compared with conventional methods (1–3) and reduce the time required for patients to receive life-saving care (4). However, TriageIntelli (5) achieved a less than 2% absolute improvement in triage accuracy using stacked machine learning models compared with base models, raising questions about whether additional complexity is clinically justified.

Previous Work:

Traditional triage is under strain from overcrowding, resource constraints and significant variation in patient prioritization due to subjective clinical assessments. Using peer-reviewed articles from PubMed, Scopus, IEEE Xplore and Google Scholar, researchers conducted a narrative review of the strengths and limitations of AI for patient triage in emergency departments. Several studies suggested that AI-driven triage may improve patient prioritization, reduce wait times and optimize resource allocation. However, key challenges were found, including data quality limitations, algorithmic bias, low trust among clinicians and overall ethical concerns. That said, the review was narrative in nature and limited to English-language articles published between 2014-2024, potentially omitting relevant findings from studies that failed to meet said criteria. (1)

Despite positive outcomes in other aspects of healthcare, the capacity of AI to assist in triaging patients in the emergency department remains relatively underexplored. Researchers conducted a search for peer-reviewed, English-language studies in the EMBASE, Ovid MEDLINE and Web of Science databases using relevant medical subject headings (MeSH) terms. The review found that, compared with conventional triage, AI integration was associated with improved triage discrimination, disease identification and resource allocation. Despite these findings, the review included several studies which used retrospective datasets from only a single teaching hospital, possibly limiting the generalisability of the results. (2)

Overcrowding within hospital emergency departments has become a major challenge, risking negative health outcomes for patients. A narrative synthesis was conducted on ten studies that evaluated AI-assisted Emergency Severity Index (ESI) triage systems involving emergency nurses as well as reporting on triage performance and challenges. AI-assisted triage enhanced patient flow as well as reducing over-triage, under-triage and patient wait times. Despite these findings, the review relied on retrospective data and included studies with limited model validation, potentially restricting real-world applicability. (3)

Based on industry guidelines, patients with ST-segment elevation myocardial infarction (STEMI) require a door-to-balloon (D2B) time of less than 90 minutes, but delays often occur due failures to interpret the electrocardiography (ECG) efficiently. An AI-based model for STEMI detection using ECG data and ASAP scores was developed using 2907 twelve-lead ECGs, including 882 STEMI and 2025 non-STEMI. Mean D2B time was reduced from 64.5 minutes to 53.2 minutes with the proportion of patients treated within 90 minutes increasing from 87.2% to 98.5%. Limitations include the study relying on a cohort of less than 200 patients from a single center. Verification of generalisability would require a larger-scale diverse study. Moreover, the study also has a false positive rate of roughly 20%, which may contribute to unnecessary prioritization of non-STEMI patients. (4)

Limited resources, overcrowding and a rising number of complex cases are putting pressure on patient triage; AI integration was identified as a means to enhance process efficiency. Seven different AI-based models were used to evaluate patient triage levels in accordance with the Korean triage and acuity score (KTAS). The Support Vector Machines (SVM) and Gradient Boosting Machines (GBM) models performed best individually with accuracies of 79% and 78.7% respectively, but the stacking model had a marginal improvement of less than 2% compared with both. However, the study used only one dataset; therefore, the model generalisability may be limited as different regions have varying clinical conditions, demographics and seasonal patterns. Real-world testing in different medical contexts is required. (5)

Inconsistent triage decisions can reduce the quality of patient care, an especially severe problem in the high-pressure environment of an emergency department. The researchers used a large language model-based intake system to obtain patient data, analysed it using a machine learning model (CatBoost) to predict triage levels. After tuning, the model achieved 85% accuracy with an F1-score of 0.960 on the ESI-1 class. Although promising, the model needs to be validated in multilingual settings to ensure fairness and generalisability. (6)

Prolonged ED stays contribute to patient flow bottlenecks which are associated with worse clinical outcomes, making accurately assigning ED disposition an operational challenge. To manage this issue, three machine learning models were trained to differentiate between those who could be discharged or admitted. The models were trained on progressively enriched feature sets: immediately apparent triage stage information from patients triaged under the Manchester Triage System (MTS), a clinical model which combined the triage stage information with four binary health indicators and a comprehensive model which combined the clinical model data with laboratory tests and diagnosis-related variables. The comprehensive model performed the best, followed by the clinical and triage-stage models. However, all three models achieved moderate predictive performance. The study was only a single-center retrospective study; furthermore, the performance of the models suffered when information could not be extracted as readily, such as in the case of another language being spoken by the patient. (7)

Emergency department personnel must rapidly determine whether patients need to be admitted on limited clinical data under rigid time and resource constraints. To address this challenge, several machine learning models were trained on triage data gathered within ten minutes of patient presentation. The models had good overall performance on hospital admission and 30-day mortality, particularly for mortality prediction. Comparatively, predictive performance on ICU admission was lower. However, the study's single-centre design may limit generalizability. (8)

Predictive models have been shown to increase triage performance but the interpretability of explainable artificial intelligence (XAI) is essential for wider clinical adoption. A retrospective, single-center study involving 351,019 emergency department patients trained and validated four models while incorporating XAI techniques. The models showed varying levels of performance, with artificial neural networks being the best. The two XAI techniques, shapley additive explanations (SHAP) and partial dependence plots (PDP), highlighted factors such as patient age and heart rate which were also used in bed allocation and patient prioritization. The study was limited as the SHAP implementation assumed feature independence which may lead to misattributed features in a real world clinical setting. (9)

Despite the promise of the technology, the use of AI for automating history-taking and triage processes has been limited. Thirteen healthcare leaders across Sweden representing seven primary care units were given semi-structured interviews. The healthcare professionals reported having issues with mistrust of the AI, adapting units to the new system of digital healthcare, and AI's functional inadequacies. However, the leaders were able to mitigate these barriers through addressing concerns, adapting units and refining the AI applications. Although rigorous, the study used snowball sampling and was unable to include a fourth Swedish region in the study due to time and resources, introducing selection bias. (10)

Deep learning algorithms have shown promise for improved patient outcomes; but, as of yet these improvements have not been fully realized in a real-world clinical setting. Thirty-eight hours of observation and seven hours of interviews were done in eleven clinics in Thailand to characterise the overall experience with a newly deployed AI-assisted eye screening. Factors such as poor lighting contributed to a ~21% image rejection rate. Further socio-environmental barriers such as network connectivity and nurses discouraging use of the system due to fear of the expensive referrals for the under-privileged were significant barriers for the deployment of the model. Moreover, the study focused on nurses and camera technicians therefore not including the patient experience in its evaluation. (11)

Accurate risk stratification and illness classification remain a key challenge in the emergency department, tasks which AI can assist in. A pre-post intervention study was done 180 days pre and post deployment of an AI-informed triage clinical decision support, measuring patient-centered performance outcomes and ED flow metrics. Post-implementation, high-acuity patient identification increased from 78.8% to 83.1% with no change in surgery hospitalisation whilst reducing times between arrival and presentation at initial care area, disposition and departure. Nurse agreement with the AI varied; it was also observed that triage performance with the system was directly proportional to their agreement with it. As the study was based on pre and post intervention, total attribution to the AI-informed triage clinical decision support cannot be guaranteed. (12)

Low-resource settings often struggle with high patient to provider ratios and no formal training in triage in areas like paediatrics; researchers proposed a human-in-the-loop AI hybrid system to address these challenges. Researchers developed a hybrid model containing a cognitive module, convolution neural network with shapley additive explanations (SHAP), and a resource-conscious optimizer. SHAP supported clinician explainability, facilitating engagement with the model's outputs. The hybrid system reduced false negatives from 12% to 5% whilst reducing the time spent by clinicians to make decisions by 31% and enhancing critical resource utilisation by 20%. The generalisability of the study is reduced by the small dataset and lack of multi-center validation. (13)

Whilst AI models show promise in mitigating crowding in emergency departments, the economic frameworks needed to evaluate these models remain comparatively underdeveloped, potentially limiting wider clinical uptake. This gap was addressed through the development of two cost-modelling frameworks and their application to the financial and operational data for three emergency departments 180 days pre and post intervention. Under the hospital management framework, which accounted for costs that were fixed, variable and modifiable, the incremental operating margin increased by US $12.6 million post-intervention, yielding a break-even threshold of US $66.02 per visit. Comparatively, the public policy approach, focusing on the cost-per-visit, found the incremental operating margin to increase at a minor US $0.9 million and break-even threshold of US $$4.69 per visit. Two significant caveats, however, are that the study focused on a singular health system and only costs specific to the emergency departments, so the findings may not apply to another hospital system or the holistic costs of a patient's medical journey. (14)

The Epic Sepsis Model (ESM) was a widely implemented sepsis early warning system in the United States; however, it had not been externally validated and largely acted as a black box model. A retrospective cohort study to externally validate the performance of ESM was conducted on 27,697 patients who had been admitted to Michigan Medicine. Researchers found that ESM showed poor discrimination ability, failing to identify 67% of patients with sepsis while also generating false alerts for 18% of hospitalizations. Still, the study was limited by being conducted in a single medical centre and its use of a composite definition of sepsis, which may reduce study generalizability. (15)

Proposed Solution:

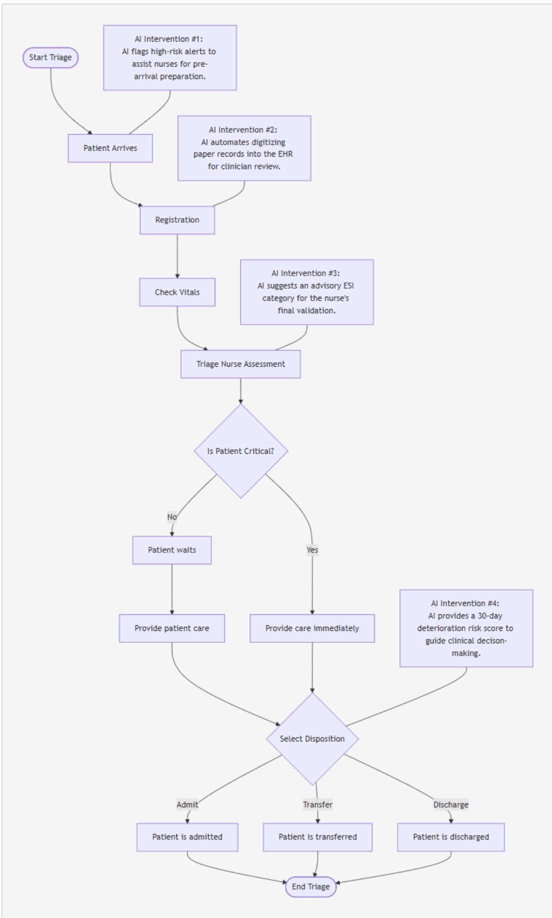
Over the course of a 12-week pilot, a stacked ensemble AI triage model will be developed and evaluated using Mercer ED data. The model performance will be compared to both the base machine learning/deep learning models and the baseline Mercer ED triage statistics. A stacked model will be considered successful if it exhibits an absolute improvement in accuracy of greater than or equal to 5% over the base models and the triage baseline. This threshold will be used to determine whether the increased complexity of a stacked ensemble model is justified for future clinical use.

Gap Analysis

However, several gaps in the literature are evident. Most studies relied on retrospective data from a single center to train the machine learning models (2,3,7). The use of this type of clinical data in model training may reduce model generalizability. Moreover, none of the reviewed models were trained on Caribbean datasets. This raises concerns regarding the applicability of these models to Caribbean populations, which may differ in comorbidity profiles and case-mix compared to the original training datasets.

Ensemble models showed improved predictive accuracy compared to the base models; however, the improvement was marginal (5). Further research is needed to determine whether ensemble approaches provide meaningful and clinically significant performance gains over simpler baseline models. This is particularly relevant due to the increased computational complexity of ensemble models where performance gains must match the additional implementation and computational costs within the resource-constrained Caribbean healthcare settings.

Workflow Diagram:



Constraints:

Constraint #1: Paper Intake Record

- Mechanism:
  Paper intake records are used which then must be transcribed digitally to be usable by an AI system.
- Design impact:
  Time and resources must be spent in order to transcribe the paper records to a digital format. This adds friction to the triage process and makes AI-based triage slower to use.
- Mitigation:
  The intake records could be done digitally in order to both speed up AI triage and provide electronic health records (EHR) with minimal extra steps. Moreover, as intake records are frequently unavailable due to another staff member using them (16), EHRs would eliminate this bottleneck by making the records always available.

Constraint #2: Lack of clinician trust

- Mechanism:
  Clinicians do not understand the AI and as such do not trust it.
- Design impact:
  AI provides a clinical judgement but clinicians may feel no reason to trust it with people's lives. This will lead to them substituting the AI's judgement for their own. The AI must then be made more trustworthy.
- Mitigation:
  AI must articulate a reason for the chosen action (i.e. choosing ESI level 5). This keeps clinicians in the loop and ensures they understand what the system is doing.

Constraint #3: Non-linear patient flow

- Mechanism:
  The triage process is inherently non-linear. Patients are often triaged and retriaged due to their evolving condition.
- Design impact:
  The AI must facilitate retriage but update the judgement based on new information. This will facilitate the non-linear nature of the emergency department.
- Mitigation:
  As the data is evolving, the triage assessment will be done on the data available at that second. If a patient deteriorates, the assessment updates to match their status.

Stakeholders:

Stakeholder #1: Sister Patrice Alleyne (ED Nurse-in-Charge)

- Concern:
  Does the AI add friction to the triage process?
- Success Definition:
  The AI can accurately make assessments within Mercer ED without requiring new long forms to fill out for each patient.

Stakeholder #2: Triage Nurse

- Concern:
  Does the AI correctly triage the patient quickly?
- Success Definition:
  The system accurately chooses an ESI level for each patient with processing times under 3 minutes (the current Mercer ED baseline).

Stakeholder #3: ED Physicians

- Concern:
  Does the AI speed up the door-to-discharge (D2D) time for a patient?
- Success Definition:
  The AI allows the physician to more quickly come to a decision for patient disposition with no risk of erroneous discharge.

Stakeholder #4: Registration Clerk

- Concern:
  Can the AI be quickly and easily incorporated into the data-entry workflow?
- Success Definition:
  The overall system can incorporate paper-based intake forms to acquire information.

Stakeholder #5: Charge Nurse

- Concern:
  Does the AI assign patients to go to the right room at the right time?
- Success Definition:
  The AI accurately assigns the correct zone for a patient

Risk analysis:

<u>Risk Memo</u>

The proposed system is viable and can meet the demands of a Caribbean ED context as long as the model is externally validated pre-deployment and properly monitored post-deployment for specific risk factors.

One major risk that must be mitigated is a potential change in clinical conditions post-deployment of the model which will contribute to confident, inaccurate triage assessments. As conditions are certain to change over time in different clinical contexts compared to the model's training data, this risk can be mitigated through the use of evaluation of model performance on a weekly basis by researchers, if an unseen trend is observed a high priority bulletin will immediately be released to clinicians to warn them of the new risk to increase vigilance. A signal of success would show itself in the form of no drops in model performance post-deployment that are sustained for 2 weeks or more.

The undertriage of critical cases is another significant risk factor as this failure can delay treatment and directly lead to worsened patient outcomes. This risk will be mitigated by making the AI more interpretable through a plain language justification of the triage result within a wider

Clinician-in-the-Loop interface; if a nurse disagrees based on their experience, they retain total authority to escalate the triage score at will. The signal of success for this risk would be a rate of false negatives for critical cases being observed to remain below 1% of all inferences made.

Training data underrepresentation is another risk that would lead to the model performing worse on populations less frequently seen in the training data. This risk will be mitigated by sourcing data from multiple emergency departments in the Caribbean such that no subgroup for each variable has fewer than 100 cases in the training data; otherwise, the model cannot be deployed until that threshold is reached. To show the successful mitigation of this risk, the model's performance for any subgroup may not fall five percentage points or more below the aggregate mean performance.

Although these risks can be thoroughly mitigated, an AI system by its nature requires significant public education to be ethically implemented as the Caribbean public, especially given their history of institutional/clinical distrust, is likely to be resistant to the idea of AI-based triage regardless of actual efficacy.

Risk Register

Risk Name	**Category**	**Likelihood**	**Impact**	**Mitigation**	**Signal of Success**
Distribution Shift	AI-technical	H. Outbreaks/Mass-casualty events/Seasonal variations in disease are statistical certainties which will differ from training data.	H. The model will confidently make inaccurate assessments on patient profiles different from those it was trained on leading to mistriage.	Developers evaluate the model's performance weekly, looking for any unseen trends in the training dataset to incorporate into retraining the model. If a new pattern is observed, a clinical bulletin is issued to alert nurses to that specific pattern and the model's apparent blindness to it.	No sustained drops in model accuracy are observed; drift alerts are monitored and responded to quickly

Undertriage of Critical Cases	AI-technical	L. Well trained models have low rates of undertriage.	H. An undertriaged critical case gets treated slower when time is critical resulting in worsened clinical outcomes.	At each inference, triage nurses examine the AI triage against their own experience, giving them the ability to escalate triage levels at will. Otherwise, the nurses may succumb to automation bias and miss a critical case.	The rate of false negatives for critical cases stays below 1% of all inferences.
EHR Failure	Operational	M. Loss of power, internet connectivity and device storage integrity are common in the Caribbean.	H. Failure of EHR records makes providing the model with patient information slower/harder.	Developers must add manual data entry options for the AI before it is deployed to the wider triage system. Otherwise, the model is not ready for deployment	Clinicians can use the AI Triage system even when EHRs are inaccessible.

Training Gap	Operational	M. Some nurses will not understand the AI at first.	H. Triage nurses unfamiliar with the program may use it incorrectly and mistriage patients	Triage nurses train with the model to accurately input patient data and evaluate model responses for correctness every 3 months. Otherwise, an uncertified nurse should use manual triage techniques until their training is complete.	90% of triage nurses can triage a patient accurately in under 2 minutes using the AI.
Consent Gaps	Ethical	H. Some patients will not want to be triaged by AI.	L. Despite obvious ethical concerns, clinical performance of the model is mostly unaffected.	Admin clerk gives every capacious patient a consent form clarifying the role of AI in their triage as well as their right to refuse. This must be done upon their arrival before the patient is triaged;	Patient consent forms satisfy GDPR/HIPAA guidelines as well as WHO principles for clinical AI.

				otherwise, the patient advocate is involved.	
Reduced Clinician Authority	Ethical	L. Clinicians already undergo extensive training making their total replacement unnecessary.	M. When totally replaced in the decision-making, clinicians are unable to override the AI and correct flawed outputs.	Developers design the model to ensure it gives an interpretable reason for a decision in plain English at every inference. Otherwise, the model is immediately recalled.	100% of inferences have an associated medical explanation for the decision available for clinicians to agree/disagree with.
Training Data Underrepresentation	Equity	M. Some groups such as those from poorer backgrounds often are neglected in clinical datasets.	H. Underrepresented groups risk being undertriaged at higher rates leading to slower medical intervention and poorer health outcomes.	Before the model is built, the developers should actively source data from a multi-centre Caribbean population containing various rare and common conditions as well as different arrival modes. No demographic class may represent less than 10% of the total	Diagnostic performance metrics for groups separated by all available classes monthly and may not fall more than 5% below the aggregate performance mean.

				training dataset, otherwise the model is not deployed.	
Unequal equipment capabilities between hospitals	Equity	H. Rural hospitals often run on equipment over a decade old.	H. Slow equipment adds friction and can increase triage time beyond clinician baseline	The inference time of the model for all hospitals will be tested by the local health authorities every 6 months. Any site where average inference time exceeds 2 minutes will be flagged for an immediate upgrade to hardware infrastructure.	100% of network hospitals maintain an average model inference time of under 2 minutes during bi-annual audits.

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